

Luka Matijević

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📄 <https://astroluka.github.io/personal-web-astroluka/>

Employment

2023 – now **Teaching Assistant**, University of Zagreb, Faculty of Science, Department of Physics, Zagreb, Croatia.

Research Interests

○ Local Universe

Environmental effects on sub-kpc galactic dynamics and morphology,
Star Formation Rate/History,
ISM/IGM properties,
ultraviolet, optical, infrared, radio data

Education

2023 – now **Doctoral study of Physics - Astrophysics**, University of Zagreb, Faculty of Science, Department of Physics, Zagreb, Croatia.

Doctoral thesis title: *"Environmental effects on the physical and chemical properties of NGC 2276 on small spatial scales"*

2017 – 2023 **Integral Undergraduate and Graduate Study of Physics**, University of Zagreb, Faculty of Science, Department of Physics, Zagreb, Croatia.

Master thesis title: *"Masses of supermassive black holes in quasars"*

Teaching Experiences

2023 – now **Teaching Assistant**, University of Zagreb, Faculty of Science, Department of Physics, Zagreb, Croatia.

- subjects: Physics 1 (Chemistry Department), Physics 2 (Mathematics Department), Introduction to Astrophysics, Physics of the Stars, Introductory Physics Lab 1

2018 – 2019 **Junior Teaching Assistant**, University of Zagreb, Faculty of Science, Department of Physics, Zagreb, Croatia.

- Assisting younger students on "Computing Laboratory" with programming tasks in C and Python programming languages

Research Experiences

2023 – now **Faculty of Science - Department of Physics, Zagreb, Croatia**, *PhD research*, under supervision of Asst. Prof. Neven Tomičić and Dr. Antonino Marasco.

- studying the influence of large-scale environmental effects on the star formation in galaxy NGC 2276 and NGC 2300 group

- 2021 **Ruđer Bošković Institute, Zagreb**, *Student Autumn Projects*, under supervision of Vibor Jelić, several months project.
 - Studying magnetohydrodynamical simulation data of interstellar medium
 - Comparison of simulation results with observational telescope data on different frequencies
- 2019 **Summer School of Science S3++**, **Požega**, *Building Your Own Plasma Speaker*, as workshop leader, one day project.
 - Teaching Summer School students how a speaker works and building a Plasma Speaker with them using microelectronic components
- 2017 **Summer School of Science S3++**, **Požega**, *Investigation of aerodynamic properties*, under supervision of Krunoslav Šešet, seven day project.
 - Learned about aerodynamics and how various car components affect aerodynamic properties
 - 3d printed two cars with different aerodynamic properties and used a wind tunnel with liquid nitrogen to test them

Work Experiences

- 2021 **Xylon d.o.o., Zagreb**, *Student Internship*, under supervision of Josip Šanko, several months project.
 - Creating Matlab-Blender interface for animations using simulated camera data on vehicles.

Finished Courses

- 2022 **The Diversity of Exoplanets**, Online course on Coursera organised by University of Geneva.
 - Gained knowledge about exoplanet formation, various methods of discovering exoplanets, exoplanetary atmospheres, etc.
- 2022 **Data-driven Astronomy**, Online course on Coursera organised by The University of Sydney.
 - Gained experience in using large datasets, creating SQL queries and machine learning methods
 - Worked with modern astronomical data such as AT20G Bright Source Sample (BSS) catalogue, SuperCOSMOS all-sky galaxy catalogue, NASA Exoplanet database and Sloan Digital Sky Survey data.
- 2022 **Illuminative**, Mobile app development course organised by EESTEC LC Zagreb.
 - Gained knowledge about creating different types of apps in Kotlin and Java programming languages.
- 2020 **Elements of AI**, Online course organised by University of Zagreb and University of Helsinki.
 - Learned about various techniques of which AI is comprised such as neural networks, machine learning, etc.
- 2020 **3D modelling course in Autodesk 3ds Max**, Online course organised by Studium College.
 - Learned to use Autodesk 3ds Max software in 3d modelling, UV unwrapping, sculpting, retopology, texturing, lighting, animation and rendering

Conferences

- Poster presentation, *"The influence of the environment on the lopsided galaxy NGC 2276"*, Multi-phase, Multi-temperature and Complex, Olbia, Italy, October 2025
- Poster presentation, *"The influence of the environment on the lopsided galaxy NGC 2276"*, CL2025: Entering a Golden Age of Galaxy Cluster Studies, Taipei, Taiwan, September 2025

- Oral presentation, *"The effects of ram pressure and tidal forces on the lopsided galaxy NGC 2276"*, Croatian Astronomical Society Scientific Meeting, Rijeka, Croatia, November 2024
- Poster presentation, *"NGC 2276 through the eyes of the Hubble Space Telescope: The effects of ram pressure and tidal forces on the various stellar and gas tracers"*, GASPIsa 2024 scientific conference, Pisa, Italy, May 2024
- Oral presentation, *"The peculiar lopsided galaxy NGC 2276"*, PhD Student Symposium, Zagreb, Croatia, April 2024

Invited Lectures

- Extragalactic meeting at the Arcetri Observatory, Florence, Italy, 2024
- *"Active galaxies and black holes"*, Observatory Zagreb, Zagreb, Croatia, 2024
- *"Active galaxies and black holes"*, Požega Gymnasium, Croatia, 2023

Successful Observing Proposals

WEAVE **Co-PI of LIFU/WEAVE proposal**, PI: Scott Trager, ID: WS2025B2-012.
title: *"The Difficult Life Of Galaxies In Groups: The Peculiar Case Of NGC 2276"*

VLA **PI of L-band, D-configuration proposal**, ID: VLA/26A-162.
title: *"Mapping the neutral gas properties in the longest ram pressure stripped tail"*

Activities

- Member of Croatian Astronomical Society (HAD), Zagreb, Croatia, 2025 -
- Member of Tongueo project developed inside Spock Incubator on Faculty of Electrical Engineering and Computing, Zagreb, Croatia, 2022
- Brainhack hackathon, Zagreb, Croatia, 2021
- Technical Assistant, Summer School of Science S3++, Požega, Croatia, 2019
- Member of Society for Out of Frame Education (Društvo za edukaciju van okvira), Zagreb, Croatia, 2018 - 2019
- Volunteering at Summer School of Science S3++, Požega, Croatia, 2018

Honors & Awards

- Project HRZZ-MOBDOK-2023-8006, Mobility project for doctoral students awarded by Croatian Science Foundation, working at the Astronomical Observatory in Padova, Italy, February-July 2025
- Municipality of Velika scholarship for students in middle and higher education, Croatia, academic year 2022/2023, 2021/2022, 2018/2019
- Student DIGI Awards as member of Tongueo project team, 2022
- STEM national scholarship for students in higher education, Croatia, academic year 2019/2020, 2017/2018

Skills

- **Programming**
Usage of Python, C and Matlab programming languages

- **3d modelling**

Usage of Blender and 3ds Max software in modelling, UV unwrapping, texturing and rendering as well as beginner animation

- **Other tools:**

MS Office, $\LaTeX$, Linux/UNIX, QtiPlot, SAOImageDS9, PyQSOFit, QFitsView, ^{3D}Barolo, Galfit, Bagpipes, SExtractor,

Languages

- **Mother tongue:** Croatian

- **Foreign Language #1:** English

(Reading: B2, Listening: B2, Speaking: B2, Writing: B2)

Other

- European driving licence for AM, B, F, G vehicle classes

References

- **Asst. Prof. Neven Tomičić**
Department of Physics, Faculty of Science, University of Zagreb, Zagreb, Croatia
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- **Dr. Antonino Marasco**
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- **Prof. Vernesa Smolčić**
Department of Physics, Faculty of Science, University of Zagreb, Zagreb, Croatia
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- **Dr. Vibor Jelić**
Ruđer Bošković Institute, Zagreb, Croatia
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- **Dr. Benedetta Vulcani**
INAF Istituto Nazionale di Astrofisica Osservatorio Astronomico di Padova, Padova, Italy
benedetta.vulcani@inaf.it

First Author Publications

- 2026 **Matijević, L.**, Tomičić, N., Marasco, A. et al., *Astronomy & Astrophysics*, in press.
"The Competing Influence of Ram Pressure and Tidal Interaction in NGC 2276"

Other Publications

- 2026 **Lassen, A. E.**, Smith, R., et al. *incl. Matijević, L.*, *Astronomy & Astrophysics*, in press.
"Distinguishing ram pressure from gravitational interactions: Applying the size-shape difference method to real galaxies"
- 2024 **Roberts, I. D.**, van Weeren, R. J., et al. *incl. Matijević, L.*, *Astronomy & Astrophysics*, 689 (2024), A22-A35.
"A 100 kpc ram pressure tail trailing the group galaxy NGC 2276"